Determination of trajectory of hook of the scissor mechanism with respect to input angle and scissor constants

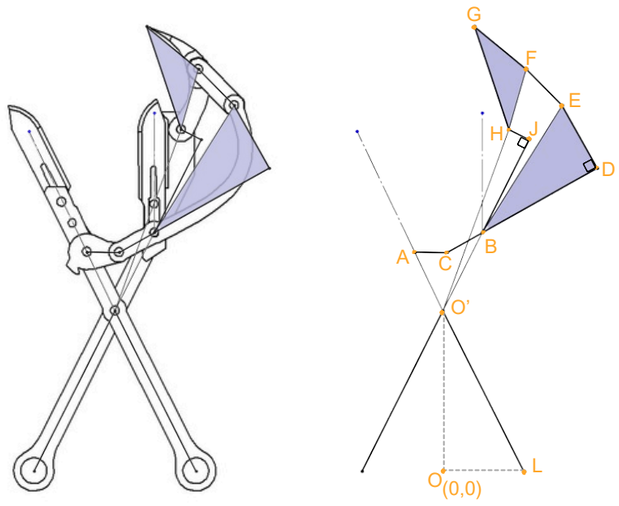


Fig 1. Kinematic representation of the mechanism

Defining of constant and variable parameters of the scissor mechanism

For the derivation of the governing equation for the trajectory of the hook (x,y) with respect to input angle θ and scissor constants, we define parameters such as -

Input angle :- θ, Output angle :- α, Trajectory points :- (x,y) [To find] and (x',y')
Scissor constants (Does not change in operation) :- Lengths - l_1, l_2, l_3, d, a_1, a_2, a_3, a_4, a_5, a_6, a_7
Scissor constants (Does not change in operation) :- Angles - λ, ψ_1, ψ_2, ψ_3
Variables during operation :- lengths - b_1, b_2, b_3
Variables during operation :- angles - Φ_1, Φ_2, Φ_3, Φ_4, Φ_5, Φ_6, Φ_7, γ_1, γ_2, γ_3, γ_4, γ_5
Origin :- O, End points :- O', A, B, C, D, E, F, G, H, J, K

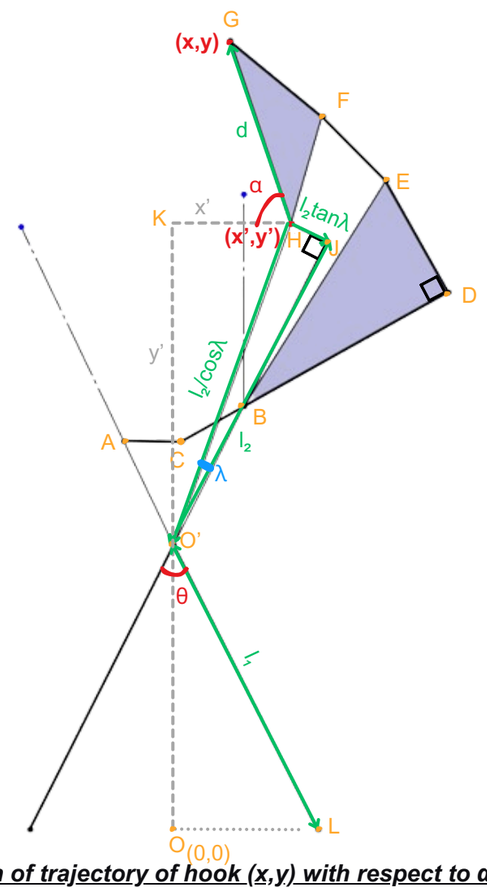


Fig 2. Determination of trajectory of hook (x,y) with respect to α and (x',y')

Derivation

According to *Fig. 2,* In △*GHK*

$$\left.\begin{aligned} x &= x' - d\cos\alpha \\ y &= y' + d\sin\alpha \end{aligned}\right\} \textit{ eq. 1}$$

Now, *KH* = x', *OK* = y', ∠*KO'H* = θ/2 - λ

$$\Rightarrow x' = \frac{l_2\sin(\theta/2-\lambda)}{\cos\lambda} \;\&\; y' = l_1\cos(\theta/2) + \frac{l_2\cos(\theta/2-\lambda)}{\cos\lambda} \quad \} \textit{ eq. 2}$$

By *eq.1* and *eq. 2* we get :-

$$x = \frac{l_2\sin(\theta/2-\lambda)}{\cos\lambda} - d\cos\alpha \;\&\; y = l_1\cos(\theta/2) + \frac{l_2\cos(\theta/2-\lambda)}{\cos\lambda} + d\sin\alpha \quad \} \textit{ eq. 3}$$

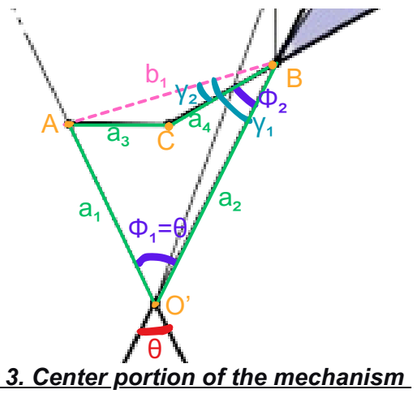


Fig 3. Center portion of the mechanism

From *Fig. 3*, $\Phi_1 = \theta$
In △*AO'B,* by cosine law -

$$b_1 = \sqrt{a_1^2 + a_2^2 - 2a_1a_2\cos\theta} \quad \} \textit{ eq. 4}$$

By sine law -

$$a_1/\sin\gamma_1 = b_1/\sin\theta$$
$$\Rightarrow \gamma_1 = \sin^{-1}(a_1\sin\theta/b_1) \quad \} \textit{ eq. 5}$$

In △*ABC*, by cosine law -

$$\gamma_2 = \cos^{-1}\left(\frac{a_4^2 + b_1^2 - a_3^2}{2a_4b_1}\right) \quad \} \textit{ eq. 6}$$

∵ $\Phi_2 = \gamma_1 - \gamma_2$, ∴ by *eq. 4, 5* & *6*

$$\Phi_2 = \sin^{-1}(a_1\sin\theta/b_1) - \cos^{-1}\left(\frac{a_4^2 + b_1^2 - a_3^2}{2a_4b_1}\right) \quad \} \textit{ eq. 7}$$

where, $b_1 = \sqrt{a_1^2 + a_2^2 - 2a_1a_2\cos\theta}$

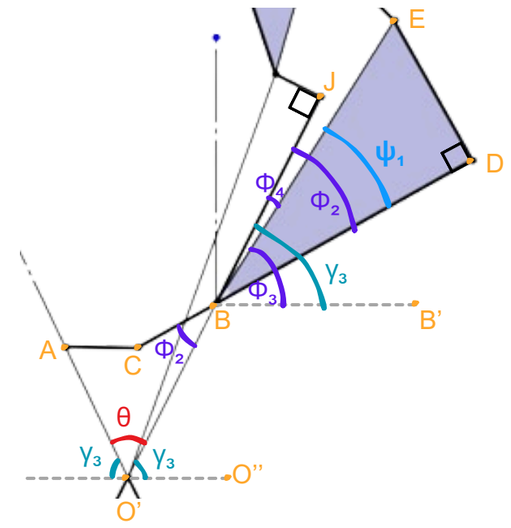


Fig 4. Right portion of the mechanism

From *Fig. 4*,

$$\left.\begin{aligned} \gamma_3 &= \pi/2 - \theta/2 \\ \Phi_4 &= \Phi_2 - \psi_1 \end{aligned}\right\} \textit{ eq. 8, 9}$$

From *eq. 9* and *eq. 7*

$$\Phi_4 = \sin^{-1}(a_1\sin\theta/b_1) - \cos^{-1}\left(\frac{a_4^2 + b_1^2 - a_3^2}{2a_4b_1}\right) - \psi_1 \quad \} \textit{ eq. 10}$$

$\Phi_3 = \gamma_3 - \Phi_4$; thus by *eq. 8* and *eq. 10*

$$\Rightarrow \Phi_3 = \pi/2 - \theta/2 + \psi_1 - \sin^{-1}(a_1\sin\theta/b_1) + \cos^{-1}\left(\frac{a_4^2 + b_1^2 - a_3^2}{2a_4b_1}\right) \} \textit{ eq. 11}$$

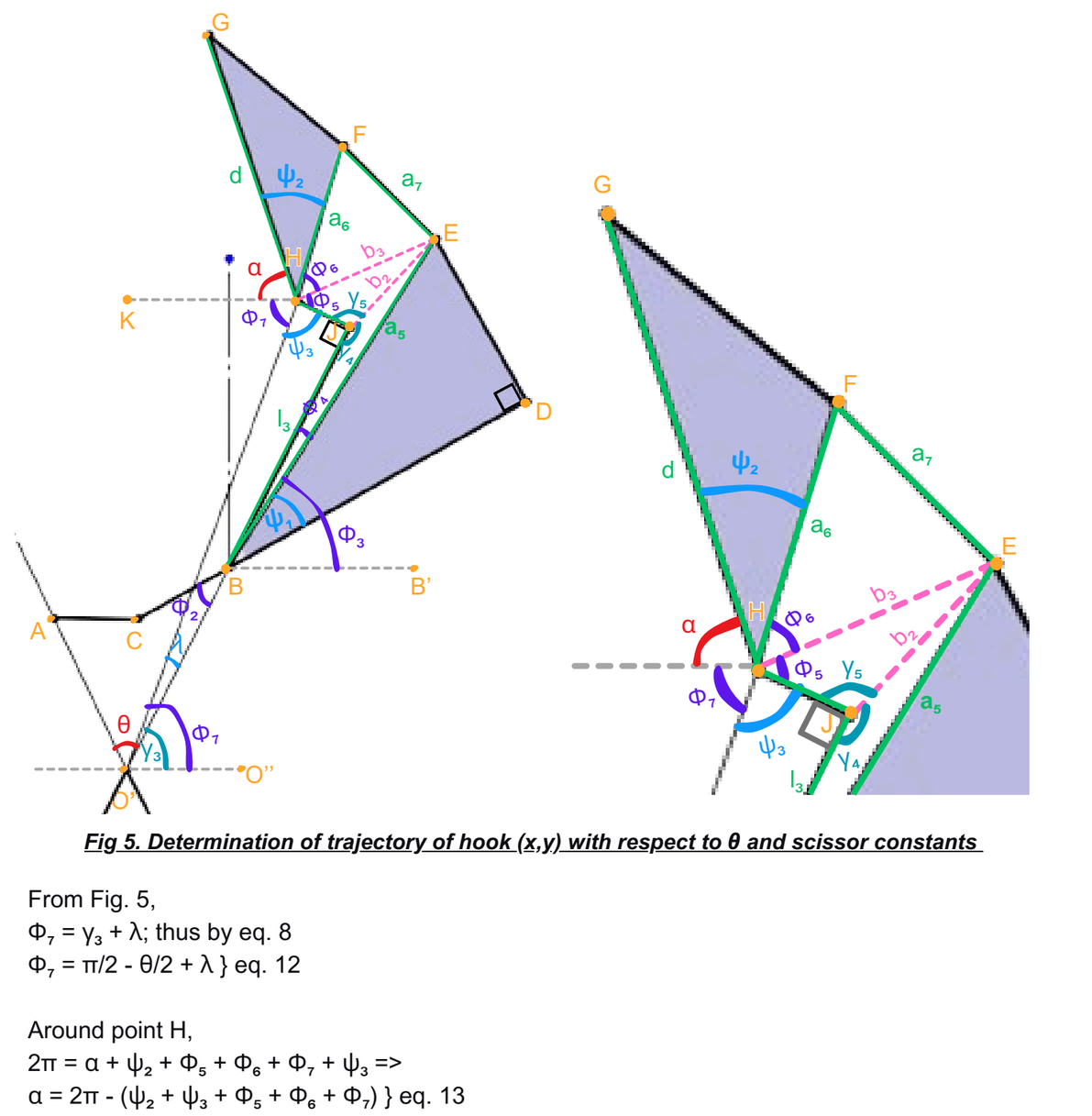


Fig 5. Determination of trajectory of hook (x,y) with respect to θ and scissor constants

From Fig. 5,

$$\Phi_7 = \gamma_3 + \lambda \text{; thus by eq. 8}$$
$$\Phi_7 = \pi/2 - \theta/2 + \lambda \} \text{ eq. 12}$$

Around point H,

$$2\pi = \alpha + \psi_2 + \Phi_5 + \Phi_6 + \Phi_7 + \psi_3 \Rightarrow$$
$$\alpha = 2\pi - (\psi_2 + \psi_3 + \Phi_5 + \Phi_6 + \Phi_7) \} \text{ eq. 13}$$

In △HJO',

$$\psi_3 = \pi/2 - \lambda \} \text{ eq. 14}$$

In △BJE, by cosine rule,

$$b_2 = \sqrt{a_5^2 + l_3^2 - 2a_5l_3\cos\Phi_4} \} \text{ eq. 15}$$

by sine rule,

$$a_5/\sin(\pi-\gamma_4) = b_2/\sin\Phi_4 \Rightarrow$$
$$\gamma_4 = \pi - \sin^{-1}(a_5\sin\Phi_4/b_2) \} \text{ eq. 16}$$

Around point *J*,

$$2\pi = \gamma_4 + \gamma_5 + \pi/2 \Rightarrow \text{by } \textit{eq. 16}$$
$$\gamma_5 = \pi/2 + \sin^{-1}(a_5\sin\Phi_4/b_2) \} \textit{ eq. 17}$$

In △*HJE*

HJ = $l_2\tan\lambda$ *(from Fig.2)* ,

by cosine law

$$b_3 = \sqrt{(l_2\tan\lambda)^2 + b_2^2 - 2{*}l_2\tan\lambda{*}b_2\cos\gamma_5} \} \textit{ eq. 18}$$

Now by cosine law again,

$$\Phi_5 = \cos^{-1}\left(\frac{(l_2\tan\lambda)^2 + b_3^2 - b_2^2}{2{*}l_2\tan\lambda{*}b_3}\right) \} \textit{ eq. 19}$$

In △*HFE*, by cosine rule

$$\Phi_6 = \cos^{-1}\left(\frac{a_6^2 + b_3^2 - a_7^2}{2a_6\,b_3}\right) \} \textit{ eq. 20}$$

∴ In *eq. 13*,

$$\alpha = 2\pi - \psi_2 - \psi_3 - \Phi_5 - \Phi_6 - \Phi_7 \quad \text{put } \textit{eq. 12, 14, 19, 20} \Rightarrow$$
$$\alpha = 2\pi - \psi_2 - \cos^{-1}\left(\frac{(l_2\tan\lambda)^2 + b_3^2 - b_2^2}{2{*}l_2\tan\lambda{*}b_3}\right) - \cos^{-1}\left(\frac{a_6^2 + b_3^2 - a_7^2}{2a_6\,b_3}\right) - \pi/2 + \theta/2 - \lambda - \pi/2 + \lambda \Rightarrow$$

$$\alpha = \pi - \psi_2 + \theta/2 + \cos^{-1}\left(\frac{(l_2\tan\lambda)^2 + b_3^2 - b_2^2}{2{*}l_2\tan\lambda{*}b_3}\right) - \cos^{-1}\left(\frac{a_6^2 + b_3^2 - a_7^2}{2a_6\,b_3}\right)$$

$$x = \frac{l_2\sin(\theta/2-\lambda)}{\cos\lambda} - d\cos\alpha \;\&\; y = l_1\cos(\theta/2) + \frac{l_2\cos(\theta/2-\lambda)}{\cos\lambda} + d\sin\alpha$$ *(eq. 3)*

where, by eq. 4, 15, 17, 18, 10 {in order}

$$b_1 = \sqrt{a_1^2 + a_2^2 - 2a_1a_2\cos\theta}$$
$$b_2 = \sqrt{a_5^2 + l_3^2 - 2a_5l_3\cos\Phi_4}$$
$$\gamma_5 = \pi/2 + \sin^{-1}(a_5\sin\Phi_4/b_2)$$
$$b_3 = \sqrt{(l_2\tan\lambda)^2 + b_2^2 - 2{*}l_2\tan\lambda{*}b_2\cos\gamma_5}$$
$$\Phi_4 = \sin^{-1}(a_1\sin\theta/b_1) - \cos^{-1}\left(\frac{a_4^2 + b_1^2 - a_3^2}{2a_4b_1}\right) - \psi_1$$

Execution of the theorem in MATLAB

θ *(input angle)* ∈ **[19.47°, 55.61°]** {*Kinematically constraint*}

Scissor constants

Input angle

Governing equation as per derivation

x vs y plotting of hook trajectory

Hook moves faster in beginning

Fig 6. MATLAB code execution

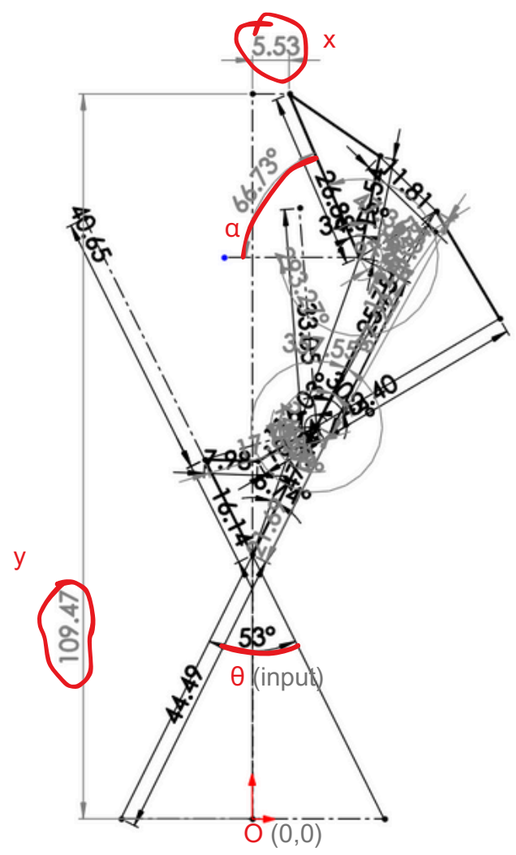


Fig 7. Kinematically constraint Sketch in SolidWorks

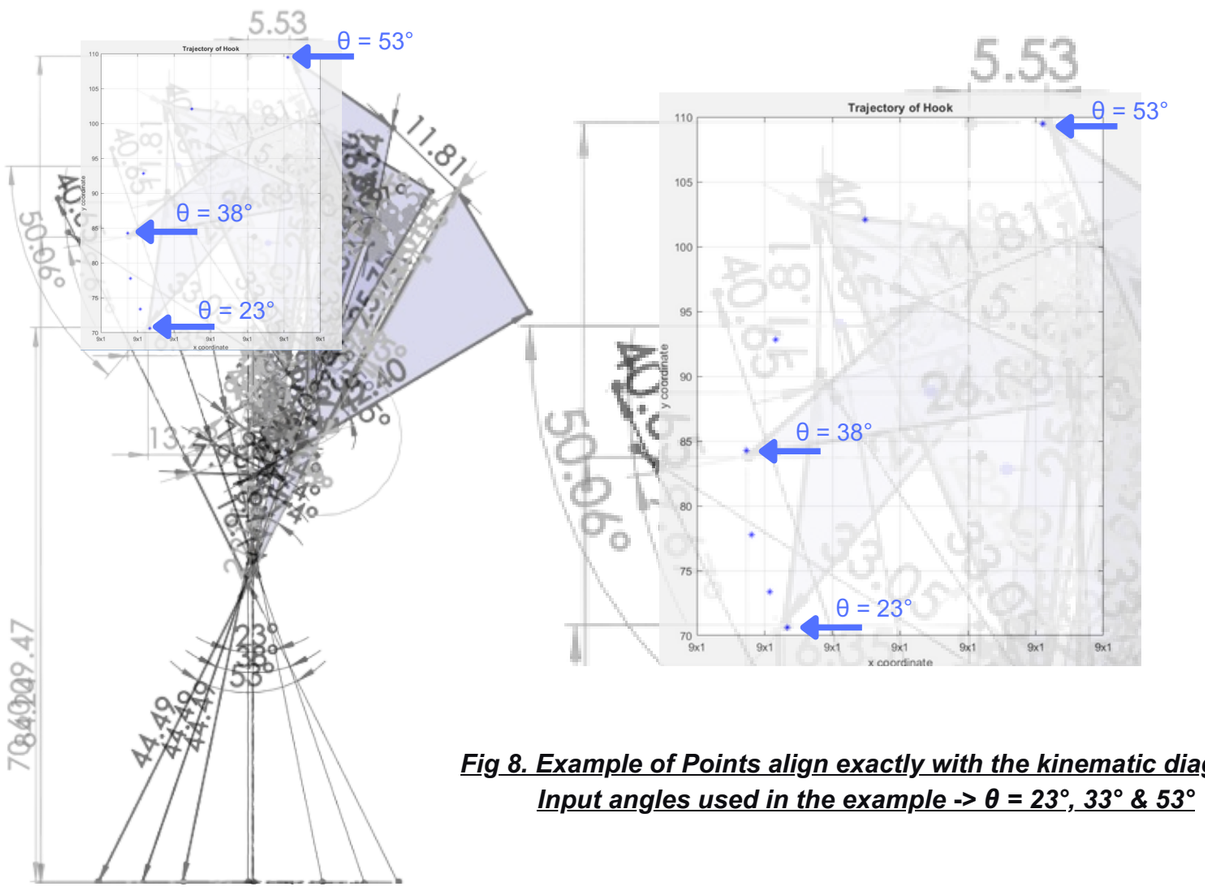
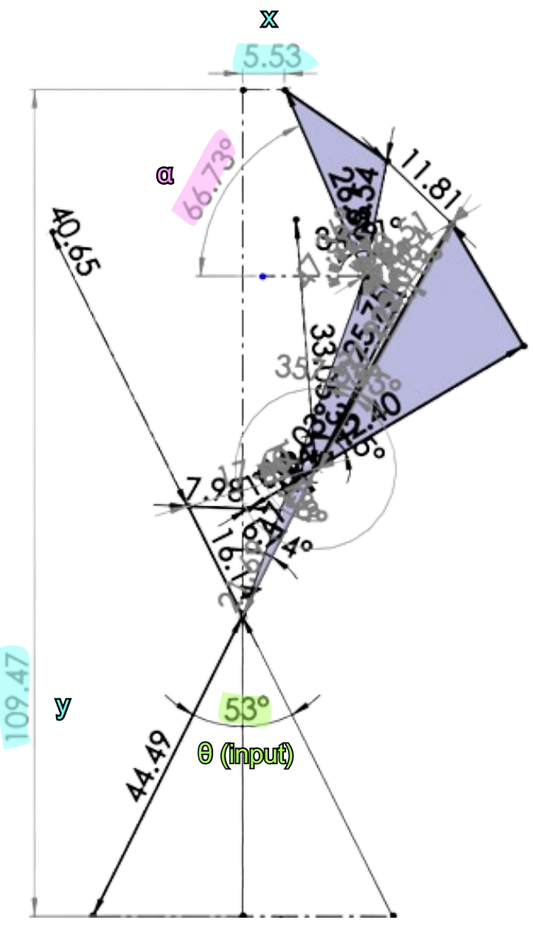


Fig 8. Example of Points align exactly with the kinematic diagram
Input angles used in the example -> θ = 23°, 33° & 53°

Name	Value
a1	16.1355
a2	21.6918
a3	7.9799
a4	10.3386
a5	37.2152
a6	15.5387
a7	11.8131
alpha	1.1648
alphadeg	66.7359
b1	17.5957
b2	11.5410
b3	13.5089
d	26.8800
gamma5	1.7092
gamma5deg	97.9278
l1	44.4941
l2	47.4422
l3	25.7504
lambda	0.1176
phi4	0.0428
phi4deg	2.4515
phi5	1.0087
phi5deg	57.7970
phi6	0.8266
phi6deg	47.3583
psi1	0.5323
psi2	0.6040
theta	0.9250
theta_deg	53
x	5.5336
y	109.4734

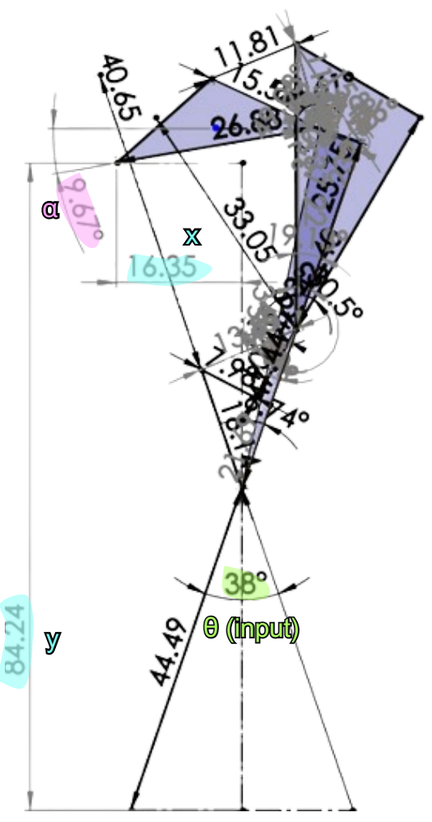
MATLAB Result



Result from
Kinematically constraint sketch

Name	Value
a1	16.1355
a2	21.6918
a3	7.9799
a4	10.3386
a5	37.2152
a6	15.5387
a7	11.8131
alpha	-0.1688
alphadeg	-9.6731
b1	13.3891
b2	15.4028
b3	11.4864
d	26.8800
gamma5	0.6573
gamma5deg	37.6582
l1	44.4941
l2	47.4422
l3	25.7504
lambda	0.1176
phi4	-0.3338
phi4deg	-19.1268
phi5	2.1815
phi5deg	124.9894
phi6	0.8565
phi6deg	49.0749
psi1	0.5323
psi2	0.6040
theta	0.6632
theta_deg	38
x	-16.3540
y	84.2365

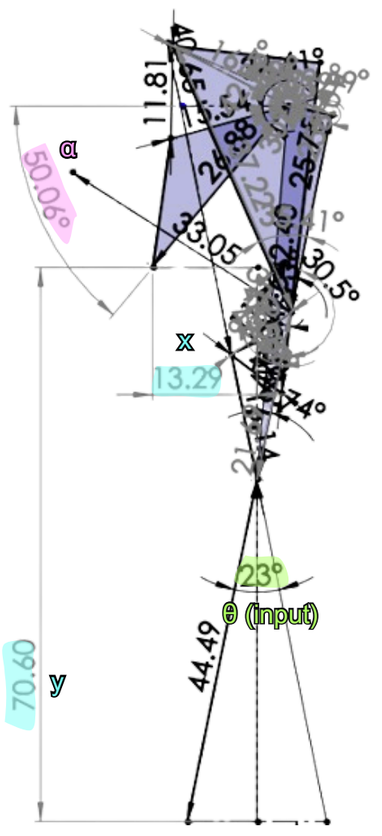
MATLAB Result



Result from
Kinematically constraint sketch

Name	Value
a1	16.1355
a2	21.6918
a3	7.9799
a4	10.3386
a5	37.2152
a6	15.5387
a7	11.8131
alpha	-0.8738
alphadeg	-50.0640
b1	9.3016
b2	22.0455
b3	16.6015
d	26.8800
gamma5	0.2093
gamma5deg	11.9911
l1	44.4941
l2	47.4422
l3	25.7504
lambda	0.1176
phi4	-0.6181
phi4deg	-35.4121
phi5	2.8621
phi5deg	163.9850
phi6	0.7500
phi6deg	42.9702
psi1	0.5323
psi2	0.6040
theta	0.4014
theta_deg	23
x	-13.2915
y	70.5980

MATLAB Result



Result from
Kinematically constraint sketch

Fig 9. Exact matching of values of the MATLAB Result and the result obtained from Kinematically constraint Sketch in SolidWorks